퀴 즈 (10)회

단원	내용	학과	학번	성명
15.6~15.7	매개변수곡면과 그 넓이 / 면적분	전전컴	2022440032	김예지

1. $y=\sin x$, $0 \le x \le \frac{\pi}{2}$를 x축 주위로 회전 시킨 곡면 S의 넓이를 구하시오.

$x=x$, $y=\sin x\cos\theta$, $z=\sin x\sin\theta$

$(0\le x\le\frac{\pi}{2},\ 0\le\theta\le 2\pi)$

$r_x=\langle 1, \cos x\cos\theta, \cos x\sin\theta\rangle$

$r_\theta=\langle 0, -\sin x\sin\theta, \sin x\cos\theta\rangle$

$r_x\times r_\theta=\langle \sin x\cos x, -\sin x\cos\theta, -\sin x\sin\theta\rangle$

$|r_x\times r_\theta|=\sqrt{\sin^2 x\cos^2 x+\sin^2 x\cos^2\theta+\sin^2 x\sin^2\theta}$

$=\sqrt{\sin^2 x\cos^2 x+\sin^2 x}$

$=\sin x\sqrt{1+\cos^2 x}$

$A=\iint_D |r_x\times r_\theta|\, dA$

$=\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\sin x\sqrt{1+\cos^2 x}\,dx\,d\theta = 2\pi\int_0^1\sqrt{1+t^2}\,dt$ //

$\cancel{(-\frac{1}{2})\cdot\frac{2}{3}(1+\cos^2 x)^{\frac{3}{2}}}$

$\cancel{\left[\frac{1}{3}(1+\cos^2 x)^{\frac{3}{2}}\right]_{\frac{\pi}{2}}^{0}=\frac{1}{3}(2\sqrt{2}-1)}$

$\boxed{\cancel{\therefore\ \frac{2\pi}{3}(2\sqrt{2}-1)}}$

$-\int_1^0\sqrt{1+t^2}$

$(1+t^2)^{\frac{3}{2}}$

$\times\frac{3}{2}\times\frac{1}{3}(1+\cos^2 x)^{\frac{1}{2}}$

2. $x^2+y^2+z^2=4(z\ge 0)$인 곡면 S에 대하여 면적분 $\iint_S \frac{z^2}{\sqrt{x^2+y^2}}dS$를 구하여라.

$x=2\sin\phi\cos\theta$, $y=2\sin\phi\sin\theta$, $z=2\cos\phi$

$(0\le\phi\le\frac{\pi}{2},\ 0\le\theta\le 2\pi)$

$w=f(x,y,z)=\frac{z^2}{\sqrt{x^2+y^2}}$

반지름이 a인 구면의 $|r_\phi\times r_\theta|=a^2\sin\phi$

$\therefore A=\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\frac{4\cos^2\phi}{\sqrt{4\sin^2\phi}}\cdot 4\sin\phi\,d\phi\,d\theta$

$=8\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\cos^2\phi\,d\phi\,d\theta$

$=8\int_0^{2\pi}\frac{\pi}{4}\,d\theta$

$=\boxed{4\pi^2}$

$\cos^2\phi=\frac{1+\cos 2\phi}{2}$

$\left[\frac{1}{2}\phi+\frac{1}{4}\sin 2\phi\right]_0^{\frac{\pi}{2}}=\frac{\pi}{4}$

$\cos^2 x=$

$\int_0^1\sqrt{1+t^2}\,dt$ 어떻게?

$\int\sqrt{1+t^2}$

$\int_0^{\frac{\pi}{4}}\frac{\sin^2\theta+\cos^2\theta}{\cos^3\theta}$

$\int_0^1\sqrt{1+t^2}\,dt$? $\int\sqrt{1+\tan^2\theta}$

$t=\tan\theta$? $dt=\sec^2\theta\,d\theta$? $\sec^3\theta$